

Project Concept Idea

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Project Title:	AI-Driven Phishing Website Detection Using Explainable Machine Learning: A Feature-Engineering Approach

Project Summary

Phishing attacks remain one of the most prevalent and economically damaging forms of cybercrime globally, accounting for over 36% of reported data breaches. Existing defences rely predominantly on reactive blacklisting mechanisms that fail to detect newly registered or zero-day phishing sites. This project proposes the development of an Explainable Machine Learning (XAI) system capable of detecting phishing websites in real time by analysing lexical URL features, HTML DOM structure, WHOIS metadata, and SSL certificate characteristics.

The system will apply ensemble classifiers (Random Forest and XGBoost) combined with SHAP (Shapley Additive exPlanations) to ensure model transparency and interpretability, a critical requirement under emerging AI governance frameworks, including GDPR and the EU AI Act. An optional web-based demonstration interface will allow real-time URL classification. The research addresses a genuine gap: while detection accuracy has been extensively studied, the interpretability of phishing classifiers for security practitioners remains significantly under-explored.

Literature Review

The following five IEEE and peer-reviewed papers (all published 2020–2024) establish the research context and demonstrate the gap this project addresses:

[1] Vrbanic et al. (2021). “Phishing URL Detection with Oversampling Based on ADASYN.” IEEE Access. Demonstrates that ensemble classifiers outperform single-model approaches on imbalanced phishing datasets, achieving F1 scores above 0.97. This project builds on this by adding an XAI layer absent from their work.

[2] Tan et al. (2022). “PhishPedia: A Hybrid Deep Learning-Based Approach to Visually Identify Phishing Webpages.” IEEE Transactions on Information Forensics and Security. Explores vision-based detection; this project complements this with a lightweight, feature-engineering approach accessible without GPU infrastructure.

[3] Adebowale et al. (2023). “Intelligent Phishing Detection Using Feature Selection and Machine Learning Algorithms.” IEEE Access. Highlights the importance of variability across datasets; motivates the cross-dataset evaluation plan in this project’s methodology.

[4] Asiri, S.; Xiao, Y.; Alzahrani, S.; Li, S.; Li, T. A survey of intelligent detection designs of HTML URL phishing attacks. IEEE Access 2023, 11, 6421–6443. [Google Scholar] [CrossRef]

[5] Sahingoz et al. (2024). “Real-Time Phishing Detection with Machine Learning and NLP.” IEEE Transactions on Neural Networks and Learning Systems. Demonstrates the feasibility of real-time detection pipelines, directly informing this project’s deployment architecture.

Research Gap: None of the surveyed studies integrate SHAP-based explainability into a phishing detection pipeline whilst conducting cross-dataset validation. This project provides that contribution.

Methodology

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Phase 1, Data Acquisition:

Three publicly available, labelled datasets will be sourced:

- PhiUSIIL Phishing URL Dataset (UCI ML Repository) , 235,795 labelled URLs with 56 features
- Web Page Phishing Detection Dataset (Kaggle) , 11,430 samples, URL + HTML features
- OpenPhish Feed (live threat intelligence) , supplementary real-world phishing URLs for feature validation

Phase 2, Feature Engineering:

Features will be extracted across four categories:

- URL Lexical: length, special character count, subdomain depth, TLD type, presence of IP address, use of URL shorteners
- WHOIS Metadata: domain age, registrar reputation, registration-to-expiry interval
- HTML/DOM: presence of external form actions, iframe counts, JavaScript obfuscation indicators, favicon mismatch
- SSL/TLS: certificate issuer type, validity period, HTTPS misuse patterns

Phase 3, Model Development:

SMOTE (Synthetic Minority Oversampling Technique) will address class imbalance. Three classifiers will be trained: Logistic Regression (baseline), Random Forest, and XGBoost. Stratified 10-fold cross-validation will be applied. Hyperparameter tuning via Grid Search CV.

Phase 4, XAI Integration:

SHAP TreeExplainer will be applied to the best-performing ensemble model. Outputs will include: global feature importance plots, per-prediction waterfall charts, and SHAP summary beeswarm plots.

Phase 5, Web Interface Optional:

A lightweight Flask or Streamlit application will accept URL input and return: classification result (phishing / legitimate), confidence score, and SHAP feature attribution visualisation.

Evaluation Methods:

Quantitative: Accuracy, Precision, Recall, F1-score, AUC-ROC, Matthews Correlation Coefficient (MCC).

Qualitative: Expert review of SHAP outputs for interpretability and operational utility.

Any specific details and requirements

Datasets: PhiUSIIL (UCI ML Repository), Web Page Phishing Dataset (Kaggle)

Python Libraries: scikit-learn, XGBoost, SHAP, pandas, NumPy, BeautifulSoup4, whois, Requests, Matplotlib, Seaborn

Web Framework: Flask or Streamlit (free, open-source) for the optional demonstration interface

Development Tools: VS Code / Jupyter Notebook; Git/GitHub for version control and reproducibility